December 2012

DIN 743-4

DIN

ICS 21.120.10

Calculation of load capacity of shafts and axles – Part 4: Fatigue limit, endurance limit – Equivalently damaging continuous stress, English translation of DIN 743-4:2012-12

Tragfähigkeitsberechnung von Wellen und Achsen –
Teil 4: Zeitfestigkeit, Dauerfestigkeit –
Schädigungsäquivalente Spannungsamplitude,
Englische Übersetzung von DIN 743-4:2012-12

Calcul de la capacité des arbres et axes –
Partie 4: Résistance à la fatigue pour une durée de vie limitée, résistance à la fatigue pour une durée de vie illimitée –
Amplitude de contrainte pour endommagement équivalent,
Traduction anglaise de DIN 743-4:2012-12

Document comprises 11 pages

Translation by DIN-Sprachendienst.
In case of doubt, the German-language original shall be considered authoritative.

© No part of this translation may be reproduced without prior permission of DIN Deutsches Institut für Normung e. V., Berlin. Beuth Verlag GmbH, 10772 Berlin, Germany, has the exclusive right of sale for German Standards (DIN-Normen).

English price group 8
www.din.de
www.beuth.de

08.15 2096511

A comma is used as the decimal marker.

Contents

Page

Foreword 3
1 Scope 4
2 Normative references 4
3 Symbols, designations and units 4
4 Calculation 5
4.1 General 5
4.2 Single direction of loading 5
4.2.1 Calculation of safety factor for time-constant amplitude in the limited life strength domain 5
4.2.2 Calculation of safety factor for load spectra in the limited life strength and fatigue strength domains 7
4.3 Combined loads 10
Bibliography 11

Figures

Figure 1 — Limited life strength value σ_{ANK} 6
Figure 2 — Method according to Miner – extended 8

Foreword

This standard has been prepared by Working Committee NA 060-34-32 AA *Wellen- und Welle-Nabe-Verbindungen* of Section *Antriebstechnik* of the *Normenausschuss Maschinenbau* (Mechanical Engineering Standards Committee) in DIN.

DIN 743, *Calculation of load capacity of shafts and axles* comprises:

— *Part 1: General*

— *Part 2: Theoretical stress concentration factors and fatigue notch factors*

— *Part 3: Strength of materials*

— *Part 4: Fatigue limit, endurance limit - Equivalently damaging continuous stress*

— *Supplement 1: Examples to part 1 to 3*

— *Supplement 2: Examples to part 4*

1 Scope

This standard applies to the calculation and verification of the safety of shafts and axles, in accordance with DIN 743-1, in the limited life strength domain (fatigue failure) as well as in the limited life strength and fatigue strength domains for load spectrum loading. Users can make use of their own verified experience or results.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

DIN 743-1, *Calculation of load capacity of shafts and axles — Part 1: General*

3 Symbols, designations and units

Symbol	Designation	Unit
D_M	Miner sum	_
i	Load bin of the load spectrum	_
j	Number of load spectrum bins	_
K_K	Load spectrum factor	_
K	Corrected number of load spectrum bins	_
n_e	Corrected number of load cycles in bin k	_
n_i	Number of load cycles in bin i	_
N_D	Salient point of the Wöhler curve between the limited life strength and fatigue strength domains	_
N_L	Required number of load cycles	_
N^*	Reference number of load cycles	_
N_S	Salient point of the Wöhler curve between the quasi-static strength and the limited life strength domain	_
q	Exponent of the Wöhler curve	_
S	Safety of the component	_
S_{min}	Required minimum safety of the component	_
ν	Shape of the load spectrum	_
$\sigma_{zd,bADK}$, τ_{tADK}	Fatigue strength value	N/mm^2
$\sigma_{zd,bANK}$, τ_{tANK}	Limited life strength value	N/mm^2

$\sigma_{zd,ba}$, τ_{ta}	Equivalent damaging stress amplitude	N/mm^2
$\sigma_{zd,bai}$, τ_{tai}	Stress amplitude of load bin i	N/mm^2
$\sigma_{zd,ba1}$, τ_{ta1}	Maximum stress amplitude of the given load spectrum	N/mm^2
S_σ, S_τ	Single safety	–

Subscripts

b	Bending
t, τ	Torsion
zd	Tension/compression
σ	Tension/compression, bending

4 Calculation

4.1 General

The method described in this standard takes into account the complete load spectrum covering the number of load cycles before the salient point N_D of the Wöhler curve. The decrease in fatigue strength caused by higher effective loads has been considered by approximation. The influence of order is not taken into account. The assumptions on which the Wöhler curve is based apply for *notched* round bars.

σ_{ADK} shall be calculated in accordance with DIN 743-1. For cases where σ_m = constant and/or τ_{tm} = constant it shall be assumed that σ_m and τ_{tm} have a constant value in all bins of the load spectrum and in the case of load increases remain constant in all bins of the load spectrum; this condition accordingly applies to $\sigma_{zd,ba}/\sigma_m$ = constant and/or τ_a/τ_m = constant, i.e. the relation of the stress amplitude to the mean stress is constant in all bins of the load spectrum.

4.2 Single direction of loading

4.2.1 Calculation of safety factor for time-constant amplitude in the limited life strength domain

4.2.1.1 Basic equations

The safety values shall be calculated on the basis of the maximum endurable stress (strength) and the stress existing in the case of application.

The calculated safety factor S shall be equal to or greater than the minimum safety factor S_{min}:

$$S_{min} \geq 1{,}2 \tag{1}$$

The principles of the calculation method itself require a minimum safety factor S_{min} = 1,2. Uncertainties in the assumed load, possible consequential damage, etc. require higher safety factors. They shall be agreed upon or determined by the designer, depending on the case of application.

4.2.1.2 Working stresses

If a calculation is to be made in the limited life strength domain without taking into consideration load spectra, the working stresses shall be determined according to DIN 743-1.

If load spectra are to be taken into consideration, proceed as described in subclause 4.2.2 of this standard.

4.2.1.3 Limited life strength value

Regarding the limited life strength domain the Wöhler curve equation, which is used to convert fatigue strength to limited life strength, generally applies:

$$\sigma_{zd,bANK} = \sqrt[q_\sigma]{\frac{N_D}{N_L}} \cdot \sigma_{zd,bADK} \tag{2}$$

where

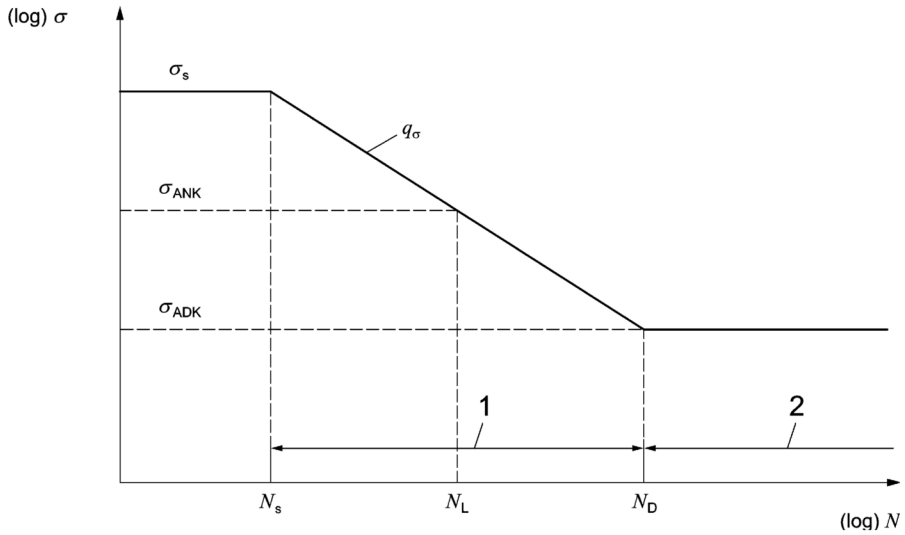
q_σ is the exponent of the Wöhler curve for bending

$$\tau_{tANK} = \sqrt[q_\tau]{\frac{N_D}{N_L}} \cdot \tau_{tADK} \tag{3}$$

where

q_τ is the exponent of the Wöhler curve for torsion.

Unless otherwise specified or known by experience, for notched round bars $q_\sigma = 5$ for bending or tension/compression and $q_\tau = 8$ for torsion can be assumed. Equations (3) and (4) apply for $N_S \leq N_L \leq N_D$. N_S and N_D are mainly dependent on σ_S, $q_{\sigma,\tau}$, σ_{AD} and $\alpha_{\sigma,\tau}$ or $ß_{\sigma,\tau}$. For the salient points of the Wöhler curve, $N_D = 10^6$ or $N_S = 10^3$ is assumed here (Figure 1). If other values are to be used they shall be set accordingly.



Key

1 Limited life strength domain
2 Fatigue strength domain

Figure 1 — Limited life strength value σ_{ANK}

4.2.2 Calculation of safety factor for load spectra in the limited life strength and fatigue strength domains

4.2.2.1 Basic equations

When calculating a safety factor by taking into consideration the effect of load spectra, the calculation procedure according to DIN 743-1 is applied.

Fatigue strength: $S = \sigma_{zd,bADK}/\sigma_{zd,ba}$ or $S = \tau_{tADK}/\tau_{ta}$ (4)

Limited life strength: $S = \sigma_{zd,bANK}/\sigma_{zd,ba}$ or $S = \tau_{tANK}/\tau_{ta}$ (5)

$$S \geq S_{min}$$

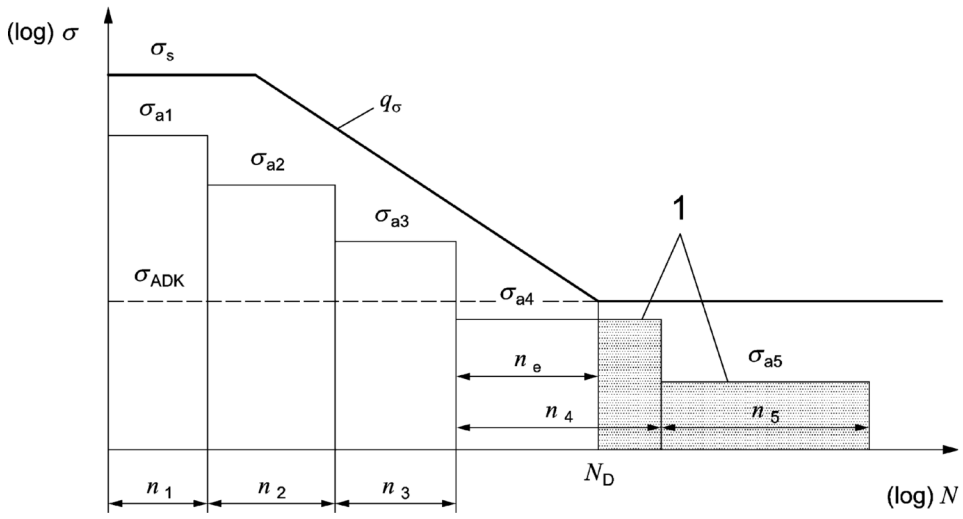
The principles of the calculation method itself require a minimum safety factor $S_{min} = 1{,}2$. Uncertainties with the assumed load, possible consequential damage, etc. require higher safety factors. They shall be agreed upon or otherwise determined.

A progression of the Wöhler curve similar to Figure 1 is prerequisite for the calculation method.

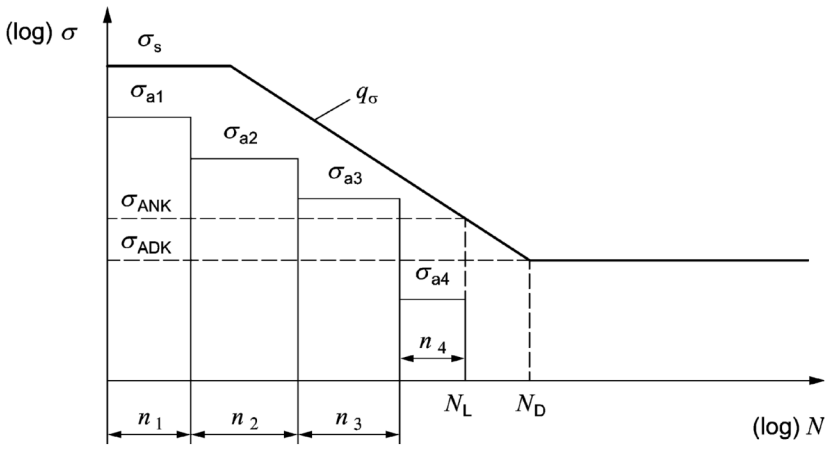
NOTE Recent tests show that in the case of very high cycle fatigue ($N > 10^8$) the fatigue strength can continue to taper off. This is mainly put down to the effect of micro inclusions. In some cases, fatigue strength also continuously tapers off with $N > N_D$. This was found, for example, on through-hardened rolling bearing steels and under the influence of aggressive media. This tapering off of fatigue strength shall be taken into consideration separately.

4.2.2.2 Working stresses

To maintain the method of proving the fatigue strength in accordance with DIN 743-1, from the given load spectrum and the (nominal) stress spectrum that can be directly calculated from it, a constant equivalent damaging stress amplitude is determined by the method according to “Miner – extended” (see Figure 2).



a) Share to be considered according to “Miner – extended” with $\Sigma n_i \geq N_D$



b) Share to be considered with $\Sigma n_i < N_D$

Key

1 Share not taken into account

Figure 2 — Method according to Miner – extended

NOTE The cumulative damage theory according to Palmgren-Miner, here abbreviated to “Miner”, is the basis of most of the methods applied for service life calculation considering the effect of cyclic, changing loads in a component. “Miner – extended” also takes into account shares of stress below fatigue strength which can contribute to total damage due to crack formation. Experience has shown that this method approximates results of exact analyses adequately for practical calculations.

4.2.2.3 Equivalent damaging stress amplitude

The equivalent damaging stress amplitude is to be calculated using Equation (6).

$$\sigma_{zd,ba} = \sigma_{zd,ba1}/K_{K\sigma} \text{ or } \tau_{ta} = \tau_{ta1}/K_{K\tau} \tag{6}$$

where $\sigma_{zd,ba1}$ or τ_{ta1} is the amplitude of that share of the spectrum representing the highest load. Here, $\sigma_{zd,ba1}$ or τ_{ta1} is a reference quantity that can be replaced by any nominal stress amplitude. The load spectrum factor $K_{K\sigma,\tau}$ is to be determined using Equation (7).

The load spectrum factor becomes:

$$K_{K\sigma,\tau} = \sqrt[q_{\sigma,\tau}]{\left(\frac{1}{\nu_{\sigma,\tau}^{q_{\sigma,\tau}}} - 1\right) \cdot D_M + 1} \tag{7}$$

D_M allows for deviations from the limit of the Miner sum. Statistical investigations have shown that D_M can be significantly smaller than 1. According to [1], [4]

$$D_M = 0{,}3 \tag{8}$$

is recommended.

Based on own experience, $D_M \neq 0{,}3$ can be chosen (in general: $0{,}3 < D_M \leq 1$). Other determinations shall be specified in the documentation (N_D, D_M, $q_{\sigma,\tau}$).

The shape of the load spectrum $v_{\sigma,\tau}$, taking into account the composition of the load spectrum and determining the equivalent damaging stress amplitude by Equation (6) or (7), is:

$$v_\sigma = \sqrt[q_\sigma]{\sum_{i=1}^{k}\left(\frac{n_i}{N^*}\right)\cdot\left(\frac{\sigma_{zd,bai}}{\sigma_{zd,ba\,1}}\right)^{q_\sigma}} \tag{9}$$

$$v_\tau = \sqrt[q_\tau]{\sum_{i=1}^{k}\left(\frac{n_i}{N^*}\right)\cdot\left(\frac{\tau_{tai}}{\tau_{ta\,1}}\right)^{q_\tau}} \tag{10}$$

a) Fatigue strength domain: Sum limit k (see Figure 2a).

It is assumed that only those load bins of the load spectrum contribute to damage, whose sum of load cycle numbers is equal to or smaller than the load cycle number N_D of the salient point of the Wöhler curve. Load bins of the load spectrum, whose sum of load cycles exceeds N_D, are cut off at N_D.

For the last bin, $i = k$, to be considered, set $n_{i=k} = n_e$ and reduce the total number of bins of the load spectrum j to the number of bins of the load spectrum k. Load bin k is the first one to exceed the salient point of the Wöhler curve or just to reach it (see fatigue strength, Fig. 2 a) or the last one, with the total sum Σn_i, before the salient point N_D of the Wöhler curve (see limited life strength, Fig. 2 b).

b) Limited life strength domain: The sum limit k is equal to the total number j of load spectrum bins $k = j$

If the total number of load cycles is in the fatigue strength domain, then:

$$\sum_{i=1}^{j} n_i \geq N_D; \qquad n_e = N_D - \sum_{i=1}^{k-1} n_i \tag{11}$$

If the total number of load cycles is in the limited life strength domain, then:

$$\sum_{i=1}^{j} n_i < N_D; \qquad n_e = n_j\,, \quad (k = j) \tag{12}$$

The following applies with respect to the reference number of load cycles N^*:

$$N^* = N_D, \qquad \text{if} \qquad \sum_{i=1}^{j} n_i \geq N_D \tag{13}$$

$$N^* = \sum_{i=1}^{j} n_i\,, \qquad \text{if} \qquad \sum_{i=1}^{j} n_i \leq N_D \tag{14}$$

where

j is the total number of load spectrum bins;

k is the number of that bin of the load spectrum meeting the two conditions

$$\sum_{i=1}^{k} n_i \geq N_D \qquad \text{and} \qquad \sum_{i=1}^{k-1} n_i < N_D$$

n_i is the total number of load cycles occurring in a bin.

4.3 Combined loads

In the case of combined loads, e.g. bending (and/or tension/compression) and torsion, the relevant safety factor is to be calculated using Equation (15). The calculation shall be based on $\sigma_{zd,bADK}$ or τ_{tADK} as a function of σ_{mv} or τ_{tmv} according to DIN 743-1 or the resulting values $\sigma_{zd,bANK}$, τ_{tANK} (Equation (2)*). The equivalent damaging stress amplitudes $\sigma_{zd,ba}$ or τ_{ta} (Equation (6)) are determined on the basis of the specified theory.

Limited life strength:

$$S = \frac{1}{\sqrt{\left(\frac{\sigma_{zda}}{\sigma_{zdANK}} + \frac{\sigma_{ba}}{\sigma_{bANK}}\right)^2 + \left(\frac{\tau_{ta}}{\tau_{tANK}}\right)^2}} \tag{15}$$

Fatigue strength:

$$S = \frac{1}{\sqrt{\left(\frac{\sigma_{zda}}{\sigma_{zdADK}} + \frac{\sigma_{ba}}{\sigma_{bADK}}\right)^2 + \left(\frac{\tau_{ta}}{\tau_{tADK}}\right)^2}} \tag{16}$$

The load cases σ_{mv} = constant (τ_{tmv} = constant) or σ_a/σ_{mv} = constant (τ_{ta}/τ_{mv} = constant) shall be used as a basis. Thus, these quantities are equal for all load bins and remain constant when the load is increased.

*) Translator's note. Should read "Equations (2) and (3)".

---,,,

Bibliography

[1] FKM-Richtlinie: *Rechnerischer Festigkeitsnachweis für Maschinenbauteile. Forschungskuratorium Maschinenbau (FKM), VDMA-Verlag Frankfurt/Main 2003*

[2] Haibach E.: *Betriebsfestigkeit, VDI Verlag 1989*

[3] Linke H.: *Stirnradverzahnungen Hanser Verlag 1996*

[4] Eulitz, K.G.: *Beurteilung der Zuverlässigkeit von Lebensdauervorhersagen nach dem Nennspannungskonzept und dem örtlichen Konzept anhand einer Sammlung von Betriebsfestigkeitsversuchen. Habilitationsschrift, TU Dresden 1999*

[5] FVA-Richtlinie R 743: *Erfassung von Lastkollektiven und Berechnung der Sicherheit im Dauer- und Zeitfestigkeitsbereich*

[6] Gudehus, H., Zenner, H.: *Leitfaden für eine Betriebsfestigkeitsrechnung, Verlag Stahleisen mbH Düsseldorf 1995*